Spectral–Spatial Enhancement and Causal Constraint for Hyperspectral Image Cross-Scene Classification

Lijia Dong, Jie Geng, *Member, IEEE*, and Wen Jiang, *Member, IEEE*

***Abstract*— Hyperspectral cross-scene classification refers to using only labeled data from the source domain (SD) in training and testing directly on the target domain (TD) dataset. However, there are differences between the reflection spectra of objects with the same category, which makes the cross-scene classification performance drop significantly. The task of single-domain generalization (SDG) has received extensive attention to solve the above problem. To address the discrepancy between source and TDs, a spatial–spectral enhancement and causal constraint network (S^2ECNet) in terms of both data enhancement and causal alignment is proposed in this article. To make up for the lack of data diversity in the SD, a generator is created in S^2ECNet to simulate the spectral deviation and spatial deviation from the TD. A causal contribution discriminator is also built in S^2ECNet to solve the data bias problem caused by direct feature alignment, which constructs causal contribution vectors from a causal perspective and uses contrastive learning to constrain category labels, extracting "potential" causal invariances from spectral and spatial domains. The cross-scene classification test is completed on the Pavia dataset, the HyRANK dataset, and the Houston dataset, and compared with some advanced multimodal methods. The experimental results demonstrate the effectiveness of the proposed network.**

***Index Terms*— Causal alignment, data augmentation, domain shift, hyperspectral cross-scene classification, single-domain generalization (SDG).**

I. Introduction

REMOTE-SENSING image classification is a meaningful research task in remote-sensing data interpretation [1], [2], [3]. Among them, hyperspectral image has rich spectral and spatial information, which has attracted a lot of attention. Hyperspectral object classification based on deep learning has been widely studied [4], [5], [6]. Most of the current classification methods work under the condition that they need a large amount of labeled data and assume that the training and testing data are independent and identically distributed. Some works use various fusion methods [7], [8], [9] to fuse multimode data to take advantage of their rich complementary information [10], [11], [12]. However, it is usually difficult to spend a lot of labor costs to label datasets in practical applications. At the same time, due to the working status of the sensor, environmental factors during data collection, and regional differences, spectral reflectance differences exist between samples of the same category. The above situation results in a distribution deviation between the training and testing sets from different regions is called domain shift. Domain shift leads to a large drop in performance when a model trained using source domain (SD) data is directly transferred to test on target domain (TD) data. With domain shift, the current work is divided into domain adaptation (DA) and domain generalization according to whether the TD data are used in training.

DA refers to using labeled data from the SD and unlabeled data from the TD when training a model. More precisely, semi-supervised DA (SDA) uses a small amount of labeled data in addition to unlabeled data from TD in training. Unsupervised DA (UDA) is defined as using only unlabeled data from SD in training. UDA is more suitable for practical applications. Most current methods aim to extract the invariant characteristics between the TD and SD as well as reduce the domain bias. Maximum mean discrepancy [13], [14] is first proposed for domain variance minimization to explicitly align distributions by mitigating domain variance. In [15], to alleviate domain shift, a domain adversarial neural network (DANN) inspired by generative adversarial networks (GANs) is proposed to learn domain-confusing features. In [16], a multirepresentation adaptive network (MRAN) is proposed, which uses multirepresentation alignment to capture different aspects of information and solve domain shift. In [17], local and global features are aligned based on the maximum interclass discrepancy, and task-relevant domain-invariant features are learned based on metaknowledge. These DA methods achieve decent performance, but it is usually laborious to obtain sample data from the TD in practical applications.

Domain generalization (DG) refers to not using data from the TD throughout the training process. According to the number of domains in the SD, DG can be considered separately as multi-DG (MDG) and single-DG (SDG). The basic idea of MDG is to generalize information extracted from multiple SDs to an unseen TD, while SDG means that the SD has only one data source, and SDG is usually more challenging. Currently, commonly used methods for DG include domain augmentation, metalearning, and invariant

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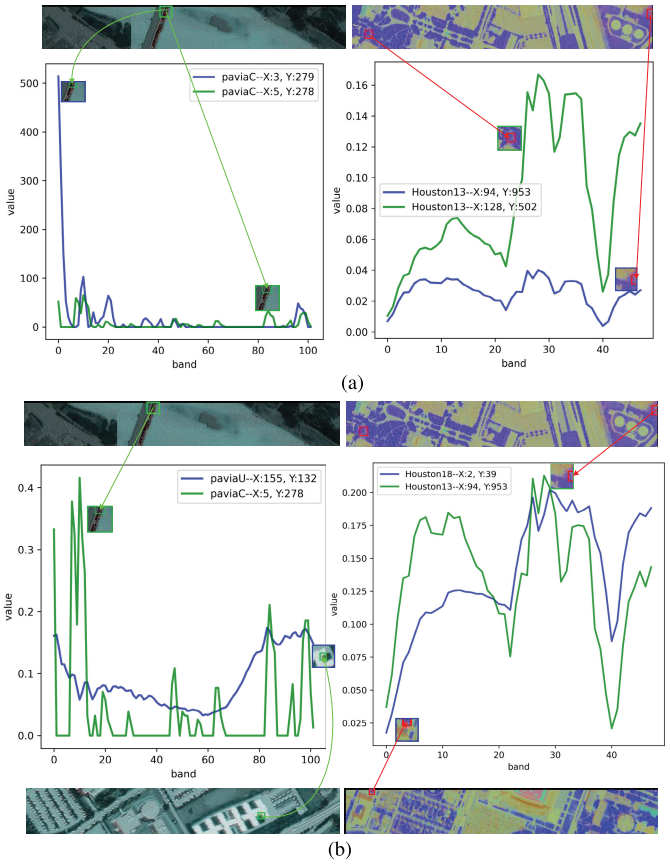


Fig. 1. Spectral curve under the specific coordinates of the hyperspectral image. (The coordinate system is shown in Figs. 7 and 8.) (a) Same category in one scene. (b) Same category in different scenes.

representation learning. Domain augmentation refers to the enhancement of data in the SD to enrich the sample diversity and improve the generalization ability of the model. In [18], higher order statistics are used to generate more diverse enhancement domains by exact histogram matching. In [19], multiple augmentations are used to build generative domains. As in [20], to simulate the diversity of the TD, metalearning refers to the division of SD data into the metatrain domain and metatest domain. In [21], a domain-augmented metalearning (DAML) method is proposed to learn generalizable representations for open domains. Extracting invariant features of the same category through certain constraints is the main idea of invariant representation learning. Currently, popular solutions to such ideas include contrastive learning and adversarial training. In [22], an adversarial local classifier is proposed as a domain discriminator, using local classification to make local domains more mixed and promote DG. In [23], to combine the advantages of the proxy-to-sample method and the sample-to-sample method, a new proxy-based contrastive learning method is proposed, which can restrain the positive alignment problem and contain rich semantic information. However, the above methods only model the statistical relationship between data and labels, build a network to extract stable features, and align sample features of the same category from the SD. As shown in Fig. 1(a), there are obvious differences between the spectrograms of data samples of the same category in the dataset, so it is difficult for the model to extract features and perform direct alignment. Meanwhile, as shown in Fig. 1(b), the spectral variability in different scenes makes it difficult to generalize the model by relying only on the regularizer to enforce feature invariance, which can lead to spurious causal features. Therefore, this article measures the causal contribution of the model under different categories from a causal perspective and extracts "potential" causal invariant features by constraining the causal contribution.

This article proposes a hyperspectral DG network based on spectral–spatial enhancement and causal constraints from a causal perspective. The network mainly includes two modules: an enhanced domain generator and a causal contribution discriminator. Compared with MDG, SDG can rely on a limited amount of domain information, and it is more difficult to generalize the extracted features to the TD that is in greater variance with the SD data. Therefore, data enhancement methods are necessary for SDG. For the characteristics of hyperspectral images, the diversity of spectral and spatial information needs to be considered simultaneously to make the generated samples effective. To address the above issues, the proposed enhanced domain generator simulates the TD's spectral and spatial sample diversity. To simulate the spectral diversity of the TD, this article utilizes the style transfer property of the adaptive instance normalization (AdaIN) layer [24] to construct a learnable spectral enhancement domain generation subnetwork to generate the spectral enhancement domain of the SD. At the same time, the ability of the spatial transformer network (STN) [25] to extract spatial rotation-invariant features is used to simulate the spatial diversity of the TD. The commonly used invariant feature extraction method is to directly align the extracted sample feature vectors. This alignment method is only a formal feature alignment between the SD data and the generated simulation data, and there is an obvious dependence on the diversity and quality of the training samples. The spatial and spectral causal contribution vectors of the SD data are calculated by the causal contribution discriminator proposed in this article using the structural causal model concept. To reduce the intraclass difference and interclass similarity, the method of contrastive learning is introduced, and the causal contribution vector is constrained according to the category label. Thereby, the "potential" causal invariant features between samples can be extracted, and the model's generalization ability can be improved.

In this article, a novel spatial–spectral enhancement and causal constraint network (S^2ECNet) is proposed for the hyperspectral cross-scene classification. The major contributions of the proposed method are summarized as follows.

1) This article introduces the concept of causal structure model for the first time in hyperspectral cross-scene classification. Aiming at the problem of limited sample diversity of SD samples and domain shift in SDG tasks, S^2ECNet addresses them from a causal perspective for hyperspectral images.
2) A causal contribution discriminator is proposed to compute spectral and spatial causal contribution vectors separately and use contrastive learning for alignment. From a causal perspective, the dependence on the diversity and quality of training samples in the direct feature

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alignment method can be alleviated, and its more "potential" causal invariant features can be extracted.

3) A spectral–spatial enhanced domain generator is proposed to simulate the sample diversity of the TD using a learnable spectral style transfer module and a spatial rotation module. Such an enhanced domain generation method is more suitable for hyperspectral images and ensures subsequent effective alignment.

The rest of this article is organized as follows. Section II describes the relevant methods for SDG and related work on causal contributions. Section III presents a new approach to SDG based on causal contribution constraints and domain enhancement. Section IV reports the experimental results. Finally, Section V briefly summarizes the conclusions of the proposed method and expectations for future work.

II. Related Works

A. Hyperspectral Image Cross-Scene Classification

The current work in hyperspectral cross-scene classification consists of two kinds, UDA and SDG, whose main difference is whether the data in the TD is seen or not during training. There is currently more working on UDA. In [26], a topological structure and semantic information transfer network (TSTnet) is proposed that uses graph convolutional networks to compensate for the limitations of CNN methods for capturing nonlocal topological relations. The graph optimal transmission is used in TSTnet for aligning topological relations, and the maximum mean difference is used to assist in distributional alignment between SD and TD. In [27], DA is combined with confidence learning by selecting high-confidence samples as training data and using adversarial learning to align the distributions of the SD and TD. In [28], multilevel DA networks are proposed to improve domain alignment of topologies using optimal transport-constrained dynamic normalization and heterogeneous knowledge graph matching, and algorithms for collaborative exploitation of data from multiple sources are proposed to improve the classification capability of the network. There is less work focusing on SDG, but due to problems such as difficulty in obtaining data in the TD or confidentiality, research in this field has important practical significance. In [29], language modality knowledge is utilized to improve the generalization ability of the model, and supervised contrastive learning is used to complete visual language alignment. However, the coarse-grained and fine-grained language modalities used in this article are difficult to obtain in most datasets.

B. Single Domain Generalization

SDG means that the SD data all come from the same domain during training. The solution of this SGD task saves the labor cost of labeling the TD dataset and improves the generality of the model in the same category of classification tasks. The SDG problem is more challenging than the MDG problem because less diverse information from the SD makes it more difficult to generalize the model to the TD.

The current common SDG solution is to use various domain enhancement methods to simulate the sample diversity of the TD and use methods, such as adversarial training and comparative learning to extract invariant features. In [30], a progressive domain expansion network (PDEN) is proposed, which uses contrastive learning to obtain domain-invariant representations by progressively generating multiple domains to simulate various transformations of photometric and geometric in unseen domains. In [31], the source distribution is enlarged via a center-aware adversarial domain augmentation method that gradually pulls the augmented samples from the respective class centroids. In [32], a set of randomly weighted shallow networks are employed to implement different appearance transformations to enhance the SD data. There is currently a little work on SDG for cross-scene hyperspectral classification. In [33], a locally linear unbiased randomization network (LLURnet) is proposed, in which style randomization is utilized to construct an extended generative network, and adversarial contrastive learning is used to obtain domain-invariant features. A domain extension network is proposed in [34], which uses an adversarial contrastive approach to learn domain-invariant features, while the network uses spatial randomization and spectral randomization to simulate SD diversity for hyperspectral images and constructs morphological convolutions in the extended domain to extract domain invariant information.

C. Causal Contribution

The causal contribution of the network refers to the influence of a feature in the neural network on the output. There are currently some methods [35], [36] that use the overall function value or perturbation with occlusion maps to evaluate the contribution of input features to the output. However, these methods do not consider the interaction between the internal parameters of the network, which leads to the deviation of the contribution measure. In [37] and [38], a causal perspective is proposed and an unbiased contribution measure called the average causal effect (ACE) is proposed. ACE uses the concept of a structural causal model, where low-level nodes interact to generate high-level nodes and uses the do(·) operation [39], [40] to represent the human intervention or perturbation on nodes. Specifically, mutual causality for input neurons in a network can be jointly caused by potential confounding factors. To explore this mutual causality, we need to compute the effect of a particular perturbation on neuron output. Therefore, the calculation process of ACE is defined as the expectation of the output class for the neuron after a specific perturbation minus the baseline expectation of the output class for the neuron plus a uniform perturbation within a fixed interval.

For the SGD problem of hyperspectral images, due to its special data properties, the "potential" impact of the spatial and spectral domains on the output is different. Therefore, it is necessary to analyze its contribution to the output as an input feature from the perspective of the spatial domain and the spectral domain, and if considered together, additional domain deviation problems will be introduced.

III. Methodology

A. Proposed Framework

This network is used to solve the cross-scene SGD problem of hyperspectral images. Annotation defines a given SD

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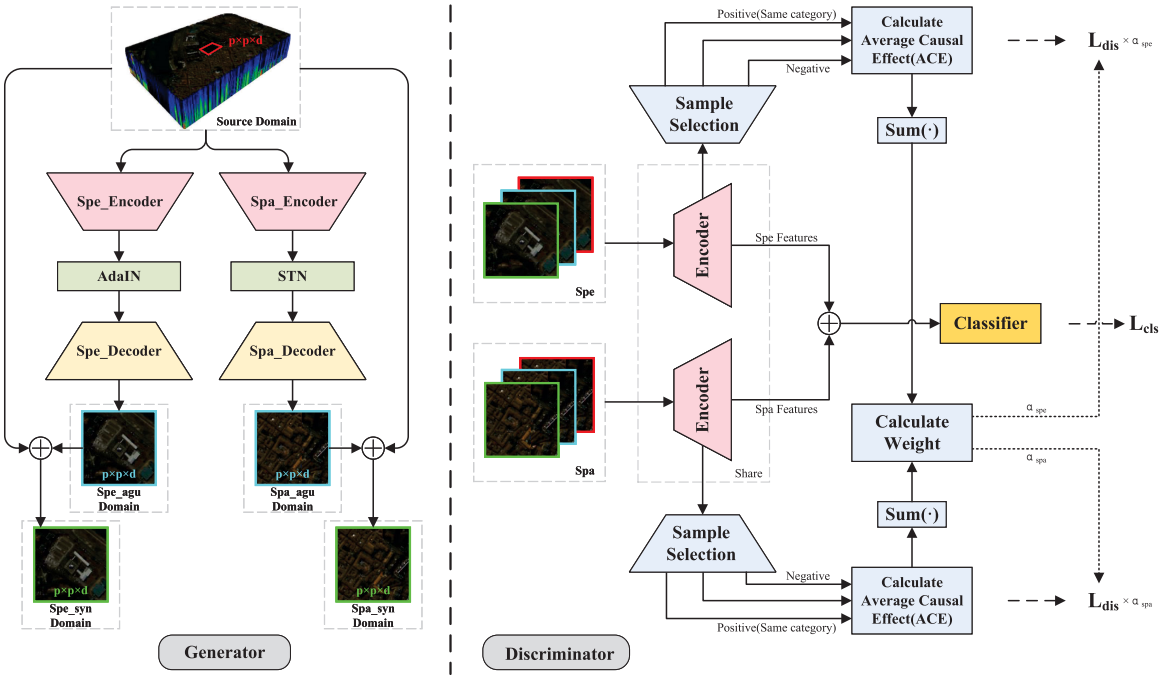


Fig. 2. Overall pipeline of the S^{2}ECNet. "⊕" in this figure means elementwise addition.

sample set as $\mathbb{S} = \{x_i^s, y_i\}_{i=1}^{N_s} \in \mathbb{R}^{d_s}$ and TD sample set as $\mathbb{T} = \{x_i^t, y_i\}_{i=1}^{N_t} \in \mathbb{R}^{d_t}$. Among them, x_i^s and x_i^t represent the SD and TD data, and y_i is the corresponding label. The number of samples and the data dimension of SD and TD are denoted by N_s and N_t, and d_s and d_t, respectively, in this task, $d_s = d_t = d$.

The SDG problem aims to train the network using just pairs of SD samples while maintaining high classification accuracy on TD. The generator and discriminator are the two fundamental components of our proposed network S^{2}ECNet, as depicted in Fig. 2. To simulate the sample diversity in the unseen TD on the SD dataset, combined with the richness of spatial and spectral information of hyperspectral images, S^{2}ECNet constructs enhanced domain generators on the spatial and spectral domains, which utilize the properties of STN and AdaIN, respectively. A pixel block from SD with the size $p \times p \times d$ serves as the network's input and is first sent to a spectral encoder (Spe_Encoder) and a spatial encoder (Spa_Encoder), where p is the specified input size. After passing the data from the SD through the encoder to generate the spectral and spatial features, the spectral features are adaptively normalized using AdaIN and then passed through the spectral decoder (Spe_Decoder) to obtain the spectral augmentation domain (Spe_AD). The spatial augmentation domain (Spa_AD) is obtained through the spatial decoder (Spa_Decoder) after using STN for spatial transformation of the spatial features. In particular, Spe_Decoder and Spa_Decoder restore features to their original size and dimensions. The obtained Spe_AD and Spa_AD are randomly weighted and summed with SD to obtain the spectral/spatial synthetic domain (Spe/Spa_YD), in case the domain offset between the randomly generated enhanced domain and the SD is too large, causing pressure on the subsequent discriminator classification and alignment.

The SD, Spa_AD, and Spa_YD are combined as the spatial group input to the discriminator. And, the SD, Spe_AD, and Spe_YD are combined as the spectral group input to the discriminator. A feature extractor with shared weights, a classifier, and a causal contribution alignment module is designed in the discriminator. The feature extractor and classifier are mainly used to predict categories and compute classification loss. The causal contribution alignment module is mainly used to measure the degree of causal contribution of sample features and impose constraints to extract causal invariant features. The classification loss is computed on the output of the classifier to improve the discriminator's ability to correctly predict the generated enhanced and synthesized domains and to ensure the reliability of the enhanced domains constructed by the generator. In the causal contribution alignment module, for the output of the features by the feature extractor, its causal effect vector for the final classification output is calculated. The causal contribution vector constructed in this article refers to the effect of the features in the neural network on the output prediction, which is described in detail in Section III-C. The basic assumption of this model is that the causal mechanism is stable across domains, and the causal alignment is accomplished by introducing a contrast loss for the causal contribution vectors, which considers the same categories across domains as positive sample pairs and different categories as negative sample pairs. The model mines the stable causal mechanisms after causal alignment, and such a modal is defined as causal invariance. Since the proposed model is constrained and aligned at the causal contribution level, forcing the feature extractor to extract features that are deterministic for category determination, such stable features are causally invariant features. Meanwhile, the idea of contrastive learning is introduced. For the current sample, positive sample pairs consist of samples from the same category,

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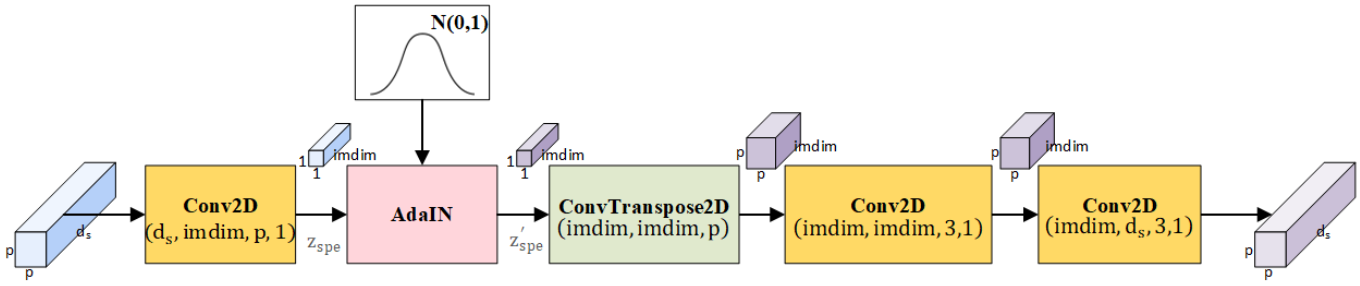


Fig. 3. Illustration of the proposed spectral enhancement domain generation subnetwork.

and negative sample pairs are constructed with the remaining samples. To ensure the effectiveness of the enhanced domain, a distance loss is designed to penalize interclass similarity and intraclass difference. The total training procedure is detailed in Algorithm 1.

Algorithm 1 Algorithm of S^{2}ECNet for UDG

Input: Source domain samples $\mathbb{S} = \{x_i^s, y_i\}_{i=1}^{N_s}$, the uniformly distributed weight w, the total epoch number T.

Output: The parameters θ_G,θ_F,θ_C of generator G, feature extractor F and classifier C.

Initialize: θ_G,θ_F,θ_C
for $epoch = 1$ **to** T **do**
  Generate augmentation and synthetic domains
  $x_{spe}^{AD}, x_{spa}^{AD} = G(x_i^s)$
  $x_{spe}^{YD} = w * x_i^s + (1-w) * x_{spe}^{AD}$
  $x_{spa}^{YD} = w * x_i^s + (1-w) * x_{spa}^{AD}$
  Calculate the causal contribution vector
  $X_{spe} = concat(x_i^s, x_{spe}^{AD}, x_{spe}^{YD})$
  $X_{spa} = concat(x_i^s, x_{spa}^{AD}, x_{spa}^{YD})$
  $t_{spe} = F(X_{spe})$
  $t_{spa} = F(X_{spa})$
  Calculate the loss L_{cls} through Eq.(8).
  Calculate the causal contribution vector c^{spe}, c^{spa} through Eq(11).
  Construct positive and negative sample pairs
  for $i = 1$ **to** N_s **do**
    $c_{pos}^{spe}/c_{pos}^{spa} = \{s|y_i = y_s, \forall s \neq i\}$
    $c_{neg}^{spe}/c_{neg}^{spa} = \{s|y_i \neq y_s, \forall s \neq i\}$
  end
  Calculate the loss L_{dis} through Eqs.(12)-(15).
  Calculate the loss L_{total} through Eq.(16).
  Update θ_F and θ_C by gradient descent
  Calculate the loss L_{cls}^{AD} through Eq.(9).
  Update θ_G by gradient descent
end

B. Enhanced Domain Generator

For the hyperspectral cross-scene SDG problem, a spectral–spatial enhanced domain generator is proposed in this article to simulate various spectral and geometric transformations in the unseen domain. The generator ensures the optimum use of the hyperspectral image's data characteristics and generates Spe_AD and Spa_AD from the perspective of spectrum and space.

1) Spectral Enhancement Domain: From the perspective of spectral characteristics, this article designs a spectral enhancement domain generation subnetwork as shown in Fig. 3 in combination with the style transfer characteristics of AdaIN.

For the input hyperspectral image block, the spectral information is first extracted using a 2-D convolution with a size of $p \times p$. At the same time, the spectral dimension is reduced to imdim to reduce network overhead, and the output is the spectral feature z_{spe} of $1 \times 1 \times$ imdim.

In this article, AdaIN is used to simulate spectral transformation in the unseen domain. AdaIN is an AdaIN layer that aligns the mean and variance of content features with the mean and variance of style features, in which the expression is as follows:

$$\text{AdaIN}_{(x,y)} = \text{FC}_1(y)\left(\frac{x-\mu(x)}{\sigma(x)}\right) + \text{FC}_2(y). \tag{1}$$

In the formula, x is the input image, y is the target conversion style image, $\mu(\cdot)$ and $\sigma(\cdot)$ refer to the mean and standard deviation, respectively, and $\text{FC}_1(\cdot)$ and $\text{FC}_2(\cdot)$ refer to the two fully connected layers.

In this article, to simulate the shift in the spectral domain, a normally distributed random number with a mean value of 0 and a variation of 1 is utilized as the target style. The specific expression is as follows:

$$z'_{\text{spe}_{(z_{\text{spe}},n)}} = \text{FC}_1(n)\left(\frac{z_{\text{spe}}-\mu(z_{\text{spe}})}{\sigma(z_{\text{spe}})}\right) + \text{FC}_2(n) \tag{2}$$

where $n \sim \mathcal{N}(0,1)$.

After obtaining the simulated spectral shift feature z'_{spe}, deconvolution is used to restore the original pixel block size $p \times p$, and a 1×1 2-D convolution is used to restore the spectral dimension so that the final output size matches the original input size, both $p \times p \times d_s$.

2) Spatial Enhancement Domain: There is also rich spatial information in hyperspectral images. Considering the spatial dimension, this article designs the spatial enhancement domain generation subnetwork as shown in Fig. 4 in combination with the spatial transformation capability of STN.

The input of this subnetwork is a pixel block of $p \times p \times d_s$, which first uses 1×1 2-D convolution to extract spatial information and compress spectral information, and the output is the spatial feature z_{spa} of $p \times p \times 3$.

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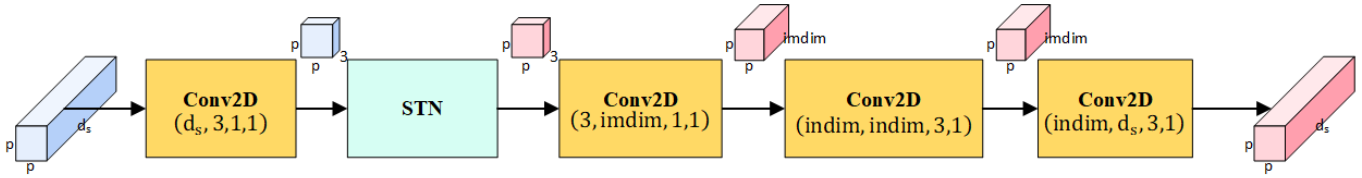


Fig. 4. Illustration of the proposed spatial enhancement domain generation subnetwork.

In this article, STN is used to simulate geometric transformations in the unseen domain. STN is a plug-and-play spatial transformer that allows active spatial transformation of the feature maps of input data in the network, thereby forcing the network to learn geometrically invariant properties. STN is mainly composed of three parts: parameter prediction (localization net), coordinate mapping (grid generator), and pixel acquisition (Sampler). According to the geometric relationship of space transformation, it can be found that no matter rotation, scaling, or translation, only the following six parameters are needed to complete:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \tau_\theta\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{bmatrix} \theta_{11} & \theta_{12} & \theta_{13} \\ \theta_{21} & \theta_{22} & \theta_{23} \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \tag{3}$$

where

$$\begin{pmatrix} x' \\ y' \end{pmatrix}$$

represents the coordinates after space transformation,

$$\begin{pmatrix} x \\ y \end{pmatrix}$$

represents the coordinates before space transformation,

$$\begin{bmatrix} \theta_{11} & \theta_{12} & \theta_{13} \\ \theta_{21} & \theta_{22} & \theta_{23} \end{bmatrix}$$

represents the affine matrix, and $\tau_\theta(\cdot)$ represents the neural network structure. This article uses the spatial feature as the input, and the output is the feature z'_{spa} that simulates the spatial offset. Specifically, the parameters in the affine matrix are obtained after the input features are passed through two convolutional layers (2-D convolutions with kernel sizes of 5×5 and 3×3) and two fully connected layers. The specific expression is as follows:

$$\begin{aligned} \begin{pmatrix} x' \\ y' \end{pmatrix}_{z'_{\text{spa}}} &= \tau_\theta\left(\left(z_{\text{spa}}\right)\right) \\ &= \begin{bmatrix} \theta_{11} & \theta_{12} & \theta_{13} \\ \theta_{21} & \theta_{22} & \theta_{23} \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}_{z_{\text{spa}}}. \end{aligned} \tag{4}$$

After obtaining z'_{spa}, 2-D convolution is used to restore the original spectral dimension d_s, so that the final output size is the same as the original input size, both $p \times p \times d_s$.

C. Causal Contribution Discriminant

There are domain shifts between data from different domains due to various reasons. A large drop in performance will happen when directly using the TD data for testing after training on the SD data. Therefore, it is particularly important to extract invariant features of the data. In the S^{2}ECNet, considering the spectral and spatial characteristics of hyperspectral images, causal invariant features are extracted from the perspective of causality in the feature and label domains. According to the above purpose, the designed causal invariant discriminator specifically includes a weight-sharing feature extractor, a classifier, and an average causal contribution alignment module.

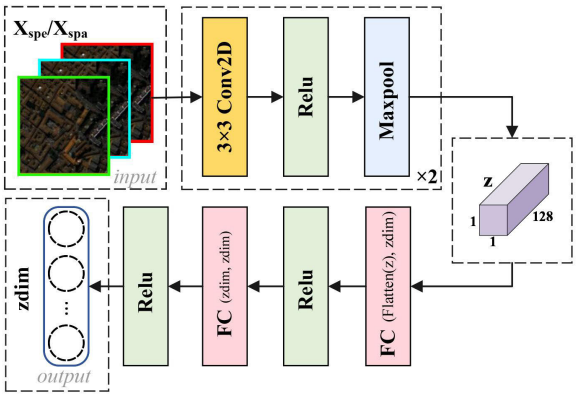


Fig. 5. Illustration of the feature extractor. "×2" in this figure indicates that the feature extraction part in the dashed box is cascaded twice.

1) Feature Extractor: The Spe_AD, Spe_YD, and SD samples generated in the generator are concatenated according to the channel dimension as the spectral dataset X_{spe}; similarly, the generated Spa_AD, Spa_YD, and SD samples are concatenated according to the channel dimension as the spatial dataset X_{spa}

$$\begin{aligned} X_{\text{spe}} &= \text{concat}\left(x_{\text{spe}}^{\text{SD}}, x_{\text{spe}}^{\text{AD}}, x_{\text{spe}}^{\text{YD}}\right) \\ X_{\text{spa}} &= \text{concat}\left(x_{\text{spa}}^{\text{SD}}, x_{\text{spa}}^{\text{AD}}, x_{\text{spa}}^{\text{YD}}\right). \end{aligned} \tag{5}$$

Here, X_{spe} and X_{spa} are sent to the feature extractor to extract the spectral feature t_{spe} and spatial feature t_{spa}. Fig. 5 depicts the construction of the feature extractor, and the extraction process is shown in the following formula:

$$\begin{aligned} t_{\text{spe}} &= F\left(X_{\text{spe}}\right) \\ t_{\text{spa}} &= F\left(X_{\text{spa}}\right). \end{aligned} \tag{6}$$

Here, $F(\cdot)$ in the formula means feature extraction. In this article, the spectral and spatial feature extractors share parameters to reduce network overhead.

2) Classifier: The spatial and spectral features extracted by the feature extractor are fed into the classifier and the prediction accuracy as the output is utilized to calculate the

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cross-entropy loss

$$l_{\text{ce}} = -\sum_{i} y_i^c \log \widehat{y}_i^c \tag{7}$$

where y_i represents the label information, $\widehat{y}_i$ represents the predicted probability, and c represents the cth class. The spectral feature t_{spe} and the spatial feature t_{spa} are added and sent to the classifier to obtain the overall classification error

$$L_{\text{cls}} = \frac{1}{N}\sum_{i} l_{\text{ce}}\left(C\left(t_{\text{spe}} + t_{\text{spa}}\right), y_i\right). \tag{8}$$

In the formula, $C(\cdot)$ represents the classifier, N represents the number of samples, and "+" represents elementwise addition. Computing the classification error ensures that the augmented domain produced by the generator is reliable. Meanwhile, to ensure that the category label of the enhanced domain is in the same domain as the SD, the classification loss of the generated enhanced domain needs to be calculated separately to update the parameters of the generator. The specific calculation method is as follows:

$$L_{\text{cls}}^{\text{AD}} = \frac{1}{N}\sum_{i} l_{\text{ce}}\left(C\left(F\left(x_{\text{spe}}^{\text{AD}}\right)\right), y_i\right) + l_{\text{ce}}\left(C\left(F\left(x_{\text{spe}}^{\text{AD}}\right)\right), y_i\right). \tag{9}$$

3) Causal Contribution Alignment Module: Considering the SGD problem from a causal perspective, there must be some kind of domain-invariant underlying causal mechanism from features to labels. The causal mechanism from features to labels for the same categories is similar but is biased for different categories. We exploit this underlying causal mechanism by quantifying and constraining the causal impact of each input feature on labels.

The ACE under the value of α is used to determine the causal contribution of neuron z^k to the output y, and the following is the specific calculation formula:

$$c^{y}_{\text{do}\left(z^k=\alpha\right)} = E\left[y|\text{do}\left(z^k=\alpha\right)\right] - E_{z'}\left[E\left[y|\text{do}\left(z^k=z'\right)\right]\right]. \tag{10}$$

In the formula, do$(\cdot)$ means a human intervention on certain variables in the system, and any value within the z^k domain can be set for α. It is assumed that z^k of the input neuron is uniformly distributed in the feature domain when there is no intervention. Therefore, the distribution of z' is $z' \sim \mathcal{U}(z_{\text{low}}^k, z_{\text{high}}^k)$, where z_{low}^k and z_{high}^k represent the minimum and maximum values in the z^k domain.

In this article, the ACE of the spectral feature t_{spe} and the spatial feature t_{spa} are calculated separately, and the calculation formula is as follows:

$$\begin{aligned} c^{\text{spe}} &= E\left[y|\text{do}\left(z^k = t_{\text{spe}}^i\right)\right] - E_{t'^i_{\text{spe}}}\left[E\left[y|\text{do}\left(z^k = t'^i_{\text{spe}}\right)\right]\right] \\ c^{\text{spa}} &= E\left[y|\text{do}\left(z^k = t_{\text{spa}}^i\right)\right] - E_{t'^i_{\text{spa}}}\left[E\left[y|\text{do}\left(z^k = t'^i_{\text{spa}}\right)\right]\right]. \end{aligned} \tag{11}$$

Among them, $t'^i_{\text{spe}} \sim \mathcal{U}(t^i_{\text{spe(low)}}, t^i_{\text{spe(high)}})$, $t'^i_{\text{spa}} \sim \mathcal{U}(t^i_{\text{spa(low)}}, t^i_{\text{spa(high)}})$, and t^i_{spe} and t^i_{spa} represent the spectral and spatial features extracted by the ith neuron in the last layer of the feature extractor, respectively. The causal contributions c^{spe} and c^{spa} of each spectral feature and spatial feature to the output label are calculated according to the above formula.

In different domains, the causal contribution of samples from the same category to the output should be consistent, and the causal contribution of samples from different categories to the output should be as different as possible, which plays a role in punishing intraclass differences and interclass similarities, so distance loss is defined as follows:

$$\begin{aligned} L_{\text{dist}}^{\text{spe}} &= \max\left\{\text{dist}\left(c_i^{\text{spe}} - c_{\text{pos}}^{\text{spe}}\right) - \text{dist}\left(c_i^{\text{spe}} - c_{\text{neg}}^{\text{spe}}\right) + \delta, 0\right\} \\ L_{\text{dist}}^{\text{spa}} &= \max\left\{\text{dist}\left(c_i^{\text{spa}} - c_{\text{pos}}^{\text{spa}}\right) - \text{dist}\left(c_i^{\text{spa}} - c_{\text{neg}}^{\text{spa}}\right) + \delta, 0\right\}. \end{aligned} \tag{12}$$

Among them, $c_{\text{pos}}^{\text{spe}}/c_{\text{pos}}^{\text{spa}}$ represents a positive sample pair, which refers to the same category, $c_{\text{neg}}^{\text{spe}}/c_{\text{neg}}^{\text{spa}}$ represents a negative sample pair, which refers to a different category, and δ is a minimum value. The distance between two features is calculated by

$$\text{dist}(x, y) = |x - y|. \tag{13}$$

There is an emphasis on spectral information and spatial information for the data characteristics of hyperspectral images. Thus, the spectral features c^{spe} and spatial c^{spa} are summed, and the proportional weight is calculated

$$\begin{aligned} \alpha_{\text{spe}} &= \frac{c^{\text{spe}}}{\text{sum}(c^{\text{spe}} + c^{\text{spa}})} \\ \alpha_{\text{spa}} &= \frac{c^{\text{spa}}}{\text{sum}(c^{\text{spe}} + c^{\text{spa}})}. \end{aligned} \tag{14}$$

A weighted sum of the distance loss for spectral features and the distance loss for spatial features is performed to calculate the total distance loss

$$L_{\text{dist}} = \alpha_{\text{spe}} \times L_{\text{dist}}^{\text{spe}} + \alpha_{\text{spa}} \times L_{\text{dist}}^{\text{spa}}. \tag{15}$$

The total distance loss L_{dist} and the classification loss L_{cls} are used to optimize the discriminator parameters to ensure the validity and reliability of the generated enhanced domain. The total loss of the network is defined in the following equation:

$$L_{\text{total}} = L_{\text{cls}} + \lambda L_{\text{dist}} \tag{16}$$

where λ is a hyperparameter for balancing the distance loss.

IV. Experiments

A. Dataset

1) Pavia Dataset: The Pavia University and Pavia Center [41], [42] datasets were acquired by the reflective optics system imaging spectrometer (ROSIS) sensors. The sensor has a total of 115 bands, and the spectral range is 430–860 nm. After processing, Pavia University data have 103 bands; Pavia Center data have 102 bands. In this article, 102 bands common to both data are used. The size of Pavia University is 610 × 340, the size of Pavia Center is 1096 × 715, and the spatial resolution is 1.3 m. Both images have nine categories of ground objects. In this article, the seven categories shared by the two data are used. The specific number of samples is shown in Table I.

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TABLE I
SAMPLE SIZE OF THE PAVIA DATASET

No.	Class	Pavia University	Pavia Center
C1	Trees	3064	7598
C2	Asphalt	6631	9248
C3	Bricks	3682	2685
C4	Bitumen	1330	7287
C5	Shadow	947	2863
C6	Meadows	18649	3090
C7	Bare soil	5029	6584
Total		39332	39355

TABLE II
SAMPLE SIZE OF THE HOUSTON DATASET

No.	Class	Houston2013	Houston2018
C1	Grass healthy	345	1353
C2	Grass stressed	365	488
C3	Trees	365	2766
C4	Water	285	22
C5	Residential buildings	319	5347
C6	Non-residential buildings	408	32459
C7	Road	443	6365
Total		2530	53200

TABLE III
SAMPLE SIZE OF THE HYRANK DATASET

No.	Class	Dioni	Loukia
C1	Dense Urban Fabric	1262	206
C2	Mineral Extraction Sites	204	54
C3	Non Irrigated Arable Land	614	426
C4	Fruit Trees	150	79
C5	Olive Groves	1768	1107
C6	Coniferous Forest	361	422
C7	Dense Sclerophyllous Vegetation	5035	2996
C8	Sparce Sclerophyllous Vegetation	6374	2361
C9	Sparsely Vegetated Areas	1754	399
C10	Rocks and Sand	492	453
C11	Water	1612	1393
C12	Coastal Water	398	421
Total		20024	10317

2) Houston Dataset: The Houston2013 [43] data were acquired by the ITRES CASI-1500 sensor and were publicly used by the IEEE GRSS Data Fusion Contest in 2013. The data size is 349×1905, and the spatial resolution is 2.5 m. The data contain 144 bands in the spectral range from 364 to 1046 nm. The ground cover is annotated into 15 categories. Houston2018 [44] is a data released during the 2018 IEEE GRSS Data Fusion competition. The data size is 2384×601, with an ultrahigh spatial resolution of 0.05 m. The data contains 48 spectral bands and 20 categories. In this article, the 48 spectral bands and seven categories shared by the two data are used, and the specific number of samples is shown in Table II.

3) HyRANK Dataset: It contains data for two regions, Dioni and Loukia [45]. The size of the Dioni is 250×1376, the size of the Loukia is 249×945, and the spatial resolution is 30 meters. The satellite hyperspectral image collected by the Hyperion sensor has 176 spectral bands and 14 consistent classes. In this article, 176 spectral bands and 12 object categories are used, and the specific number of samples is shown in Table III.

B. Implementation Details

The experimental environment is based on the Pytorch framework [46], using the Adam optimizer, and the learning rate is set to 0.001. The batch size sets on the Pavia dataset, Houston dataset and HyRANK dataset are 256, 128, and 128, respectively. We use each class's accuracy, overall accuracy (OA), average accuracy (AA), and Kappa coefficient for quantitative research.

The input of S^{2}ECNet is a pixel block with a size of patch size, and the patch size determines how much relevant information is input. We compare different patch sizes, other parameters are fixed, and different patches are selected from $\{11 \times 11, 13 \times 13, 15 \times 15, 17 \times 17\}$ for verification. Fig. 6(a) shows the impact of different patch sizes on OA on different datasets. Considering the experimental results, the patch sizes of S^{2}ECNet on the Pavia dataset, the Houston dataset, and the HyRANK dataset are selected as 13.

In the enhanced domain generation module of S^{2}ECNet, the spectral/spatial synthetic domain is obtained by the random weighted sum of the SD data and the spectral/spatial enhanced domain, and the random weighted parameters here use a uniform distribution. For the Pavia dataset, we use 50% as the training set and 50% as the validation set, and for the Houston and HyRANK datasets, we use 80% as the training set and 20% as the validation set. Furthermore, the Houston dataset is augmented by a factor of 4 with random flips and random radiation noise at training time, while the other two datasets are not augmented.

The causal contribution alignment module of S^{2}ECNet is to align the causal contribution vectors of the previous layer of the classifier. We conduct quantitative experiments on the dimension (zdim in Fig. 5) size selection of this layer, other parameters are fixed, and different zdims are selected from $\{32, 64, 128, 256\}$ for verification. Fig. 6(b) shows the effect of different zdim sizes on OA on different datasets. According to the accuracy of the classification results, the zdim size used by S^{2}ECNet is 256.

The parameter λ is an important hyperparameter in the proposed model, which determines the contribution of distance loss caused by the causal constraints. We select different λ from $\{1e-3, 1e-2, 1e-1, 1e+0, 1e+1\}$ for verification. Fig. 6(c) shows the changing trend of the classification accuracy (OA) of different λ on the three experimental datasets. It can be seen from the results that the proposed parameters can achieve better classification performance in the $1e-1$ to $1e+1$ interval. Therefore, the size of λ used by S^{2}ECNet is $1e+1$.

C. Experimental Results

We select some DA and DG methods as comparative experiments to verify the effectiveness of the proposed S^{2}ECNet. Among them, dynamic adversarial adaptation network (DAAN) [47], MRAN [16], deep subdomain sdaption network (DSAN) [48], and TSTnet [26] are DA methods. The selected DG methods are PDEN [30], learning to diversify for single domain generalization (LDSDG) [19], SagNet [49], LLURnet [33], and SDEnet [34]. In the DA method, labeled SD data (80% for training and 20% for validation) and unlabeled TD data are used. For MRAN and DSAN, due to the calculation of loss, the batch size is 32, while it is 128 in the other methods. In the DG method, only labeled SD data (80% for training and d 20% for verification) is used. Among them, PDEN, LDSDG, and SagNet's patch sizes are set at 32 due to the feature extractor's restriction on the input image

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TABLE IV

QUANTITATIVE CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE PAVIA DATASET (TD: PAVIA CENTER)

No.	Class	DAAN [47]	MRAN [16]	DSAN [48]	PDEN [30]	LDSDG [19]	SagNet [49]	TSTnet [26]	SDEnet [34]	LLURnet [33]	S^{2}ECNet
C1	Trees	72.99	67.53	88.01	77.09	92.84	98.6	64.64	89.93	90.07	84.54
C2	Asphalt	81.48	85.5	82.3	67.95	62.01	74.71	88.8	81.22	88.66	83.62
C3	Bricks	40.31	38.4	80.34	32.92	13.26	9.87	32.7	72.77	85.47	87.56
C4	Bitumen	64.11	45.07	76.11	65.16	83.71	61.05	59.38	82.54	80.40	82.01
C5	Shadow	65.1	50.45	81.63	72.51	61.54	92.42	89.66	84.81	79.36	74.71
C6	Meadows	84.55	87.92	79.51	63.33	57.02	49.03	91.91	75.11	71.39	80.23
C7	Bare soil	68.1	75.1	67.83	54.86	65.78	89.06	94.33	78.74	75.24	85.91
	OA (%)	70.63	67.24	79.43	64.59	68.86	74.04	76.09	81.94	83.08	**83.23**
	AA (%)	68.09	64.28	79.39	61.94	62.31	67.82	74.45	80.73	81.52	**82.65**
	Kappa ×100	64.81	60.67	75.43	57.27	62.42	68.54	71.47	78.33	79.65	**79.95**

TABLE V

QUANTITATIVE CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HOUSTON DATASET (TD: HOUSTON2018)

No.	Class	DAAN [47]	MRAN [16]	DSAN [48]	PDEN [30]	LDSDG [19]	SagNet [49]	TSTnet [26]	SDEnet [34]	LLURnet [33]	S^{2}ECNet
C1	Grass healthy	32.52	41.02	60.62	36.73	35.48	15.23	85.03	46.05	31.19	49.82
C2	Grass stressed	60.56	76.94	78.82	55.24	55.87	50.9	68.64	61.89	80.36	61.07
C3	Trees	60.65	65.91	68.21	42.91	43.75	62.04	52.96	60.85	56.04	66.16
C4	Water	54.55	100	100	77.27	68.18	81.82	95.45	100	100	100
C5	Residential buildings	46.37	36.9	52.72	51.26	51.62	77.03	57.13	80.61	77.76	71.03
C6	Non-residential buildings	61.26	82.68	77.44	86.16	86.14	91.24	86.37	86.42	83.12	89.45
C7	Road	61.01	56.43	46.21	27.95	28.04	5.69	37.93	42.48	48.63	48.37
	OA (%)	58.91	72.11	72.53	69.34	69.44	72.41	75.21	75.98	75.48	**77.86**
	AA (%)	53.85	65.7	69.15	53.93	52.73	54.85	69.07	68.33	68.16	**69.41**
	Kappa ×100	41.71	55.03	57.93	48.08	48.29	47.9	59.96	58.83	59.31	**61.48**

TABLE VI

QUANTITATIVE CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HYRANK DATASET (TD: LOUKIA)

No.	Class	DAAN [47]	MRAN [16]	DSAN [48]	PDEN [30]	LDSDG [19]	SagNet [49]	TSTnet [26]	SDEnet [34]	LLURnet [33]	S^{2}ECNet
C1	Dense Urban Fabric	29.06	0	5.39	0	1.46	11.17	11.65	29.13	47.09	26.7
C2	Mineral Extraction Sites	42.59	59.26	42.59	0	0	0	5.56	35.19	0	31.48
C3	Non Irrigated Arable Land	35.31	7.51	41.04	13.38	0	27	7.04	44.6	57.04	44.6
C4	Fruit Trees	11.39	12.66	36.71	0	0	0	0	5.06	3.8	32.91
C5	Olive Groves	10.09	0.9	17.58	29.36	0.36	0	10.57	1.72	0.18	0.09
C6	Coniferous Forest	16.91	37.44	33.41	0.47	0	35.07	0	38.15	24.17	46.68
C7	Dense Sclerophyllous Vegetation	59.1	67.19	72.11	85.11	86.32	79.37	80.67	77	73.56	80.67
C8	Sparce Sclerophyllous Vegetation	65.37	90.41	56.36	50.36	37.36	62.52	74.5	72.43	72.55	69.63
C9	Sparsely Vegetated Areas	53.92	5.04	30.56	70.18	73.68	30.83	71.93	36.59	22.31	54.64
C10	Rocks and Sand	0.22	2.88	28.54	13.91	51.21	0	2.43	0	0	0
C11	Water	100	100	99.64	100	100	100	100	99.93	100	100
C12	Coastal Water	100	99.76	100	85.04	99.29	100	99.76	100	100	100
	OA (%)	55.93	60.45	59.36	60.26	56.33	58.9	62.62	62.31	60.74	**63.77**
	AA (%)	43.66	40.25	47	37.31	37.47	37.16	38.68	44.98	41.73	**48.95**
	Kappa ×100	46.3	49.76	50.47	50.48	44.48	48.1	52.93	52.95	51.22	**55.77**

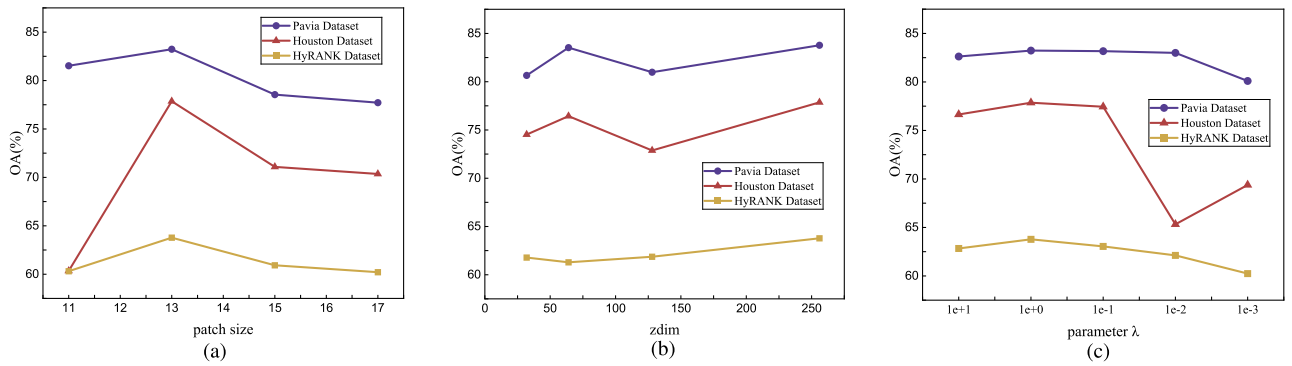


Fig. 6. OA (%) of different parameter selections on the three datasets. (a) Patch size. (b) zdim. (c) λ.

size. For the Pavia dataset, select Pavia University as the SD and Pavia Center as the TD. For the Houston dataset, choose Houston2013 as the SD and Houston2018 as the TD. For the HyRANK dataset, Dioni is chosen as the SD and Loukia as the TD. Tables IV–VI show the results of various categories, OA, AA, and Kappa on the Pavia, Houston, and HyRANK datasets, respectively, and the best results are bolded. Figs. 7–9 show the visual classification results of different models on the three datasets.

From the experimental results, the DA and DG methods have their advantages on different datasets. On the Pavia dataset, the OA of DAAN is 3%–5% higher than that of PDEN and LDSDG. However, DAAN does not perform well on the Houston dataset and the HyRANK dataset. In particular, SDEnet is the only proposed method for the hyperspectral cross-scene SDG problem, so compared with other methods, it has achieved good results on the three datasets. The S^{2}ECNet proposed in this article for the hyperspectral SDG problem

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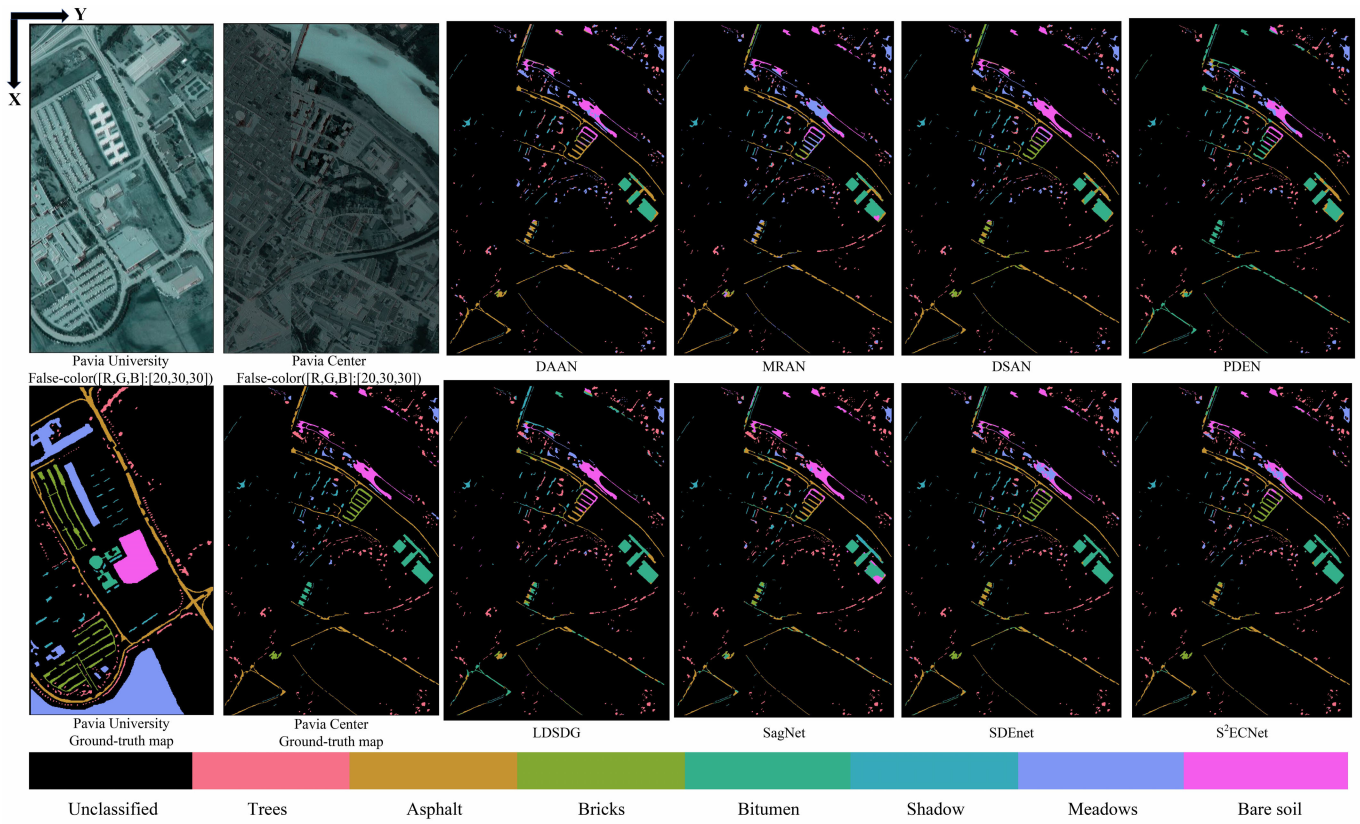


Fig. 7. Visualization and classification maps of different models on the Pavia dataset (TD: Pavia Center).

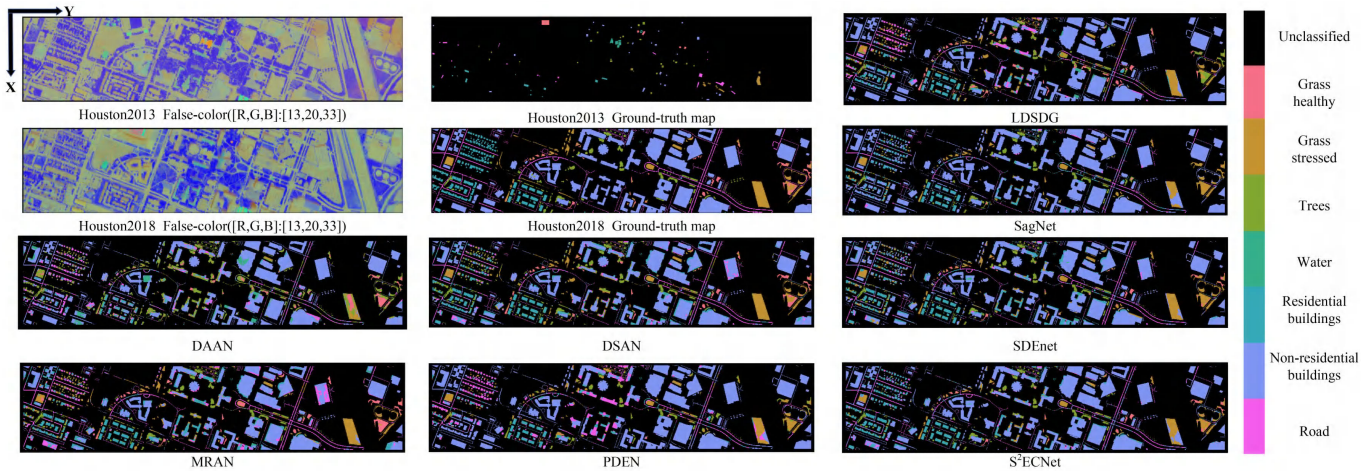


Fig. 8. Visualization and classification maps of different models on the Houston dataset (TD: Houston2018).

achieves the best OA, AA, and Kappa, which demonstrates the effectiveness of the method.

The DA methods use unlabeled samples from the TD, but the experimental results show that their classification effect is worse than that of S^{2}ECNet. This can demonstrate the effectiveness of the enhanced domain generator designed for the hyperspectral data characteristics, which can effectively simulate the data characteristics from the unseen domain. For the problem of direct alignment, S^{2}ECNet has achieved good classification results for the shadow class in the Pavia dataset and the Grass stressed class in the Houston dataset compared with other DG methods. At the same time, S^{2}ECNet has obtained the highest AA and Kappa, indicating that the method is more robust.

To compare the computational complexity of different methods, we calculate the running time of different comparative experiments in testing and training for one epoch. The experimental results (unit: seconds) are shown in Table VII. For the fairness of the experiments, all the experiments are carried out under the same hardware and software settings. Since the causal contribution vector of features needs to be calculated in S^{2}ECNet, the running time will be longer than

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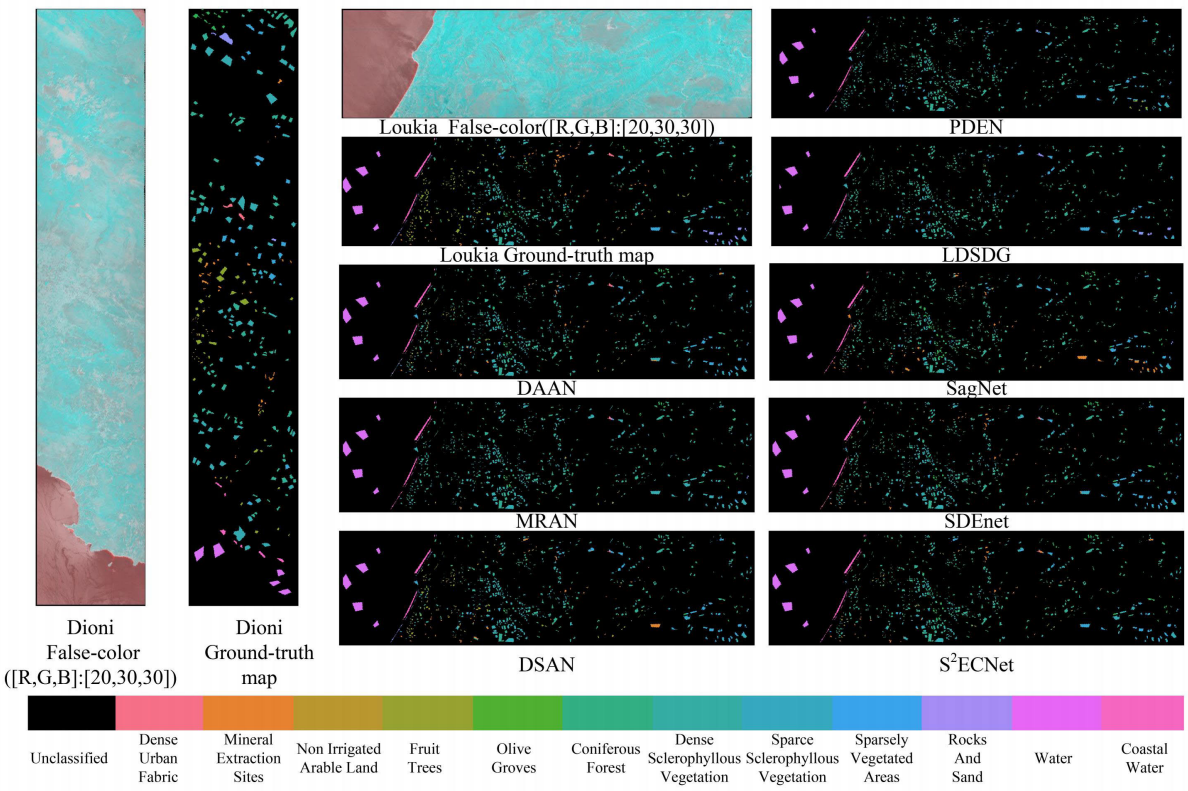


Fig. 9. Visualization and classification maps of different models on the HyRANK dataset (TD: Loukia).

other directly aligned DG methods. However, due to the simplicity of the model, S^{2}ECNet requires very little testing time.

D. Ablation Studies

To verify the effectiveness of the proposed model in S^{2}ECNet, we design a series of ablation experiments on the Pavia dataset to demonstrate the effects of the proposed enhanced domain generation module and causal contribution alignment module, respectively. The results of the ablation experiments are shown in Table VIII. Specifically, "femb" means detection on the TD directly after training on SD data. "no AdaIN" means to remove the spectral enhancement part in the enhancement domain generation module, leaving only the spatial enhancement part. "no STN" means to remove the spatial enhancement part in the enhancement domain generation module, leaving only the spectral enhancement part. "no ACE" means removing the causal contribution alignment module, and directly aligning the spectral/spatial features t_{spe}/t_{spa} obtained by 6.

According to the ablation experiment results, it can be seen that the OA of "femb" has decreased by 6.62% compared with the proposed S^{2}ECNet, indicating that directly applying the model trained with SD data to the TD will cause a huge drop in classification accuracy. This further illustrates the importance of research on hyperspectral SDG methods. Compared with S^{2}ECNet, the OA of "no AdaIN" and "no STN" has decreased by 3.23% and 1.55%, respectively, which proves the effectiveness of the enhanced domain generation module for spectral/spatial simulation of hyperspectral data characteristics. The OA of "no ACE" has decreased by 4.4% compared with S^{2}ECNet, which proves that although the direct alignment method has improved compared with "femb," it cannot obtain "potential" causality. This further demonstrates the effectiveness of the proposed causal contribution alignment module.

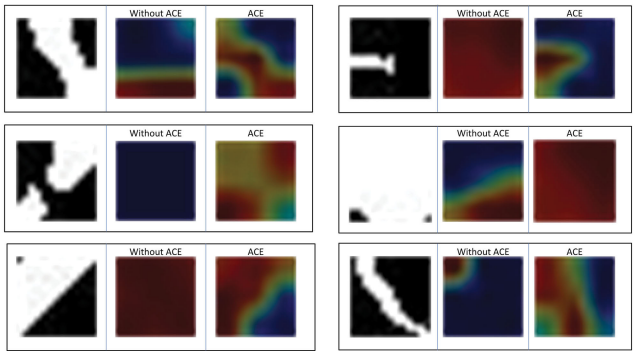


Fig. 10. Visualization of attention maps of the last convolutional layer on Pavia dataset, with the Pavia Center as the unseen TD. Every three images in the figure are a group. The binary image is a ground-truth map consistent with the center pixel label of each patch. (The white area is the same category.) "Without ACE" means that S^{2}ECNet removes the causal contributions module, and uses the category direct alignment method. "ACE" indicates that S^{2}ECNet includes the causal contribution alignment module.

To further verify that the causal contribution alignment module can capture the effective information of the samples, Fig. 10 shows the feature heat map obtained resulting from the two-layer convolution of the feature extractor. The white area of the leftmost binary image in each group of images represents the pixels consistent with the center pixel

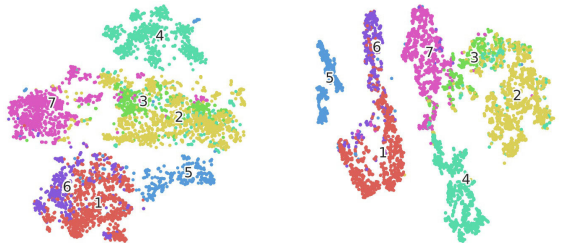
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TABLE VII
RUNNING TIME OF DIFFERENT METHODS ON THREE DATASETS (SECONDS)

No.	Class	DAAN [47]	MRAN [16]	DSAN [48]	PDEN [30]	LDSDG [19]	SagNet [49]	SDEnet [34]	S^{2}ECNet
Pavia	Training	170.34	169.14	138.63	158.12	326.94	188.77	56.97	314.04
	Testing	110.56	225.07	172.51	272.66	244.26	274.03	50.38	50.21
Houston	Training	40.06	38.3	43.13	28.01	56.88	37.68	36.98	86.48
	Testing	32.75	49.02	26.08	31.79	38.33	61.14	13.11	13.62
HyRANK	Training	82.62	80.08	88.97	126.51	122.02	127.9	45.07	271.33
	Testing	47.98	68.92	41.53	145.54	97.31	140.16	22.89	22.65

TABLE VIII
EXPERIMENTAL RESULTS OF MODULE ABLATION ON THE PAVIA DATASET

No.	Class	femb	no AdaIN	no STN	no ACE	S^{2}ECNet
C1	Trees	85.77	80.3	90.5	89.51	84.54
C2	Asphalt	85.65	77.55	81.96	80.35	83.62
C3	Bricks	49.57	86.52	66.44	76.42	87.56
C4	Bitumen	63.66	85.18	85.89	80.84	82.01
C5	Shadow	80.82	80.23	89.42	76.74	74.71
C6	Meadows	71.75	78.35	72.59	74.05	80.23
C7	Baresoil	79.16	75.38	73.59	66.25	85.91
	OA	76.61	80	81.68	78.83	**83.23**
	AA	73.77	80.5	80.06	77.74	**82.65**
	Kappa	71.82	76.21	78.04	74.65	**79.95**



(a) (b)

Fig. 11. Class separability of the proposed S^{2}ECNet using the Pavia dataset. (a) Original TD sample distribution and (b) feature distribution obtained after S^{2}ECNet alignment.

category, and the black is others. For each patch, pixels that are consistent with the central pixel category should be given greater attention. From the visualization results, more accurate pixel attention can be obtained from the perspective of causality. Fig. 11 shows the category distribution of the test set data before and after alignment. It can be seen that S^{2}ECNet can effectively separate categories.

V. CONCLUSION

In this article, a novel S^{2}ECNet is proposed for hyperspectral cross-scene classification. Specifically, S^{2}ECNet includes an enhanced domain generation module and a causal contribution alignment module. The enhanced domain generator simulates the spectral and spatial sample diversity of TD, improving the effectiveness of model generalization. The causal contribution discriminator adopts the concept of structural causal models to extract spectral and spatial causal contribution vectors. Furthermore, the causal invariant features are learned using contrastive learning. We design comparative experiments on the Pavia dataset, the Houston dataset, and the HyRANK dataset, and the proposed method achieves the best results among state-of-the-art methods. Moreover, a series of ablation experiments on the Pavia dataset demonstrate that the proposed module can effectively simulate the sample diversity of the TD and extract the "latent" causal invariant properties of samples of the same category.

The current model achieves outstanding performance in hyperspectral cross-scene classification. However, the existing hyperspectral cross-domain datasets generally have data imbalance problems. Meanwhile, the indiscriminate selection of data in the SD in S^2ECNet makes the labeled samples in the seen domain underutilized. In future work, we will combine active learning to observe the improvement of model generalization ability through data selection.

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